

The background image shows a person's hand typing on a laptop keyboard. Overlaid on the image is a semi-transparent dashboard with various data visualizations. The dashboard includes a line graph titled 'Performance of Devices' showing data points over time, a bar chart titled 'Cost of Energy' showing energy costs, and a line graph titled 'Grid Capacity' showing capacity over time. There are also some numerical readouts and a sidebar with navigation options.

MSAN Smart Power Monitoring Solution

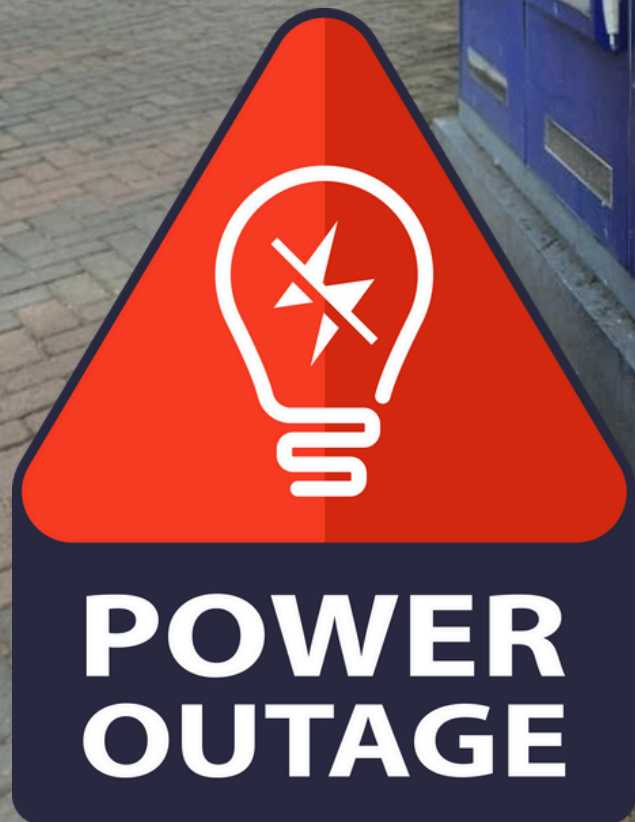
Optimizing fault detection and maintenance with intelligent IoT.

Introduction

When MSAN lost power. It will send an alarm to the NOC saying “**AC Mains Failure**”

There can be few reasons that result AC Mains Failure

- 1) CEB power loss
- 2) CEB low voltage
- 3) Breaker turned off (Main, Trip, Auto Reclosure, MSAN IN)



✧ AI-generated content



Problem

If AC Main Failure dose not resolve within MSAN's battery backup time, Technical team has to **visit that MSAN** and confirm the situation

Problem is they **can't identify the exact reason** until they visit the MSAN

this will result **unnecessary visits** to MSANs. Specially in the hard to reach areas like Kandy.

Solution

We can **install IoT device to monitor** these conditions

- 1) CEB Power presence
- 2) MSAN input voltage
- 3) Breaker Status

By looking at the **online dashboard**.
Regional office can decide whether they want to send a technical team there or not



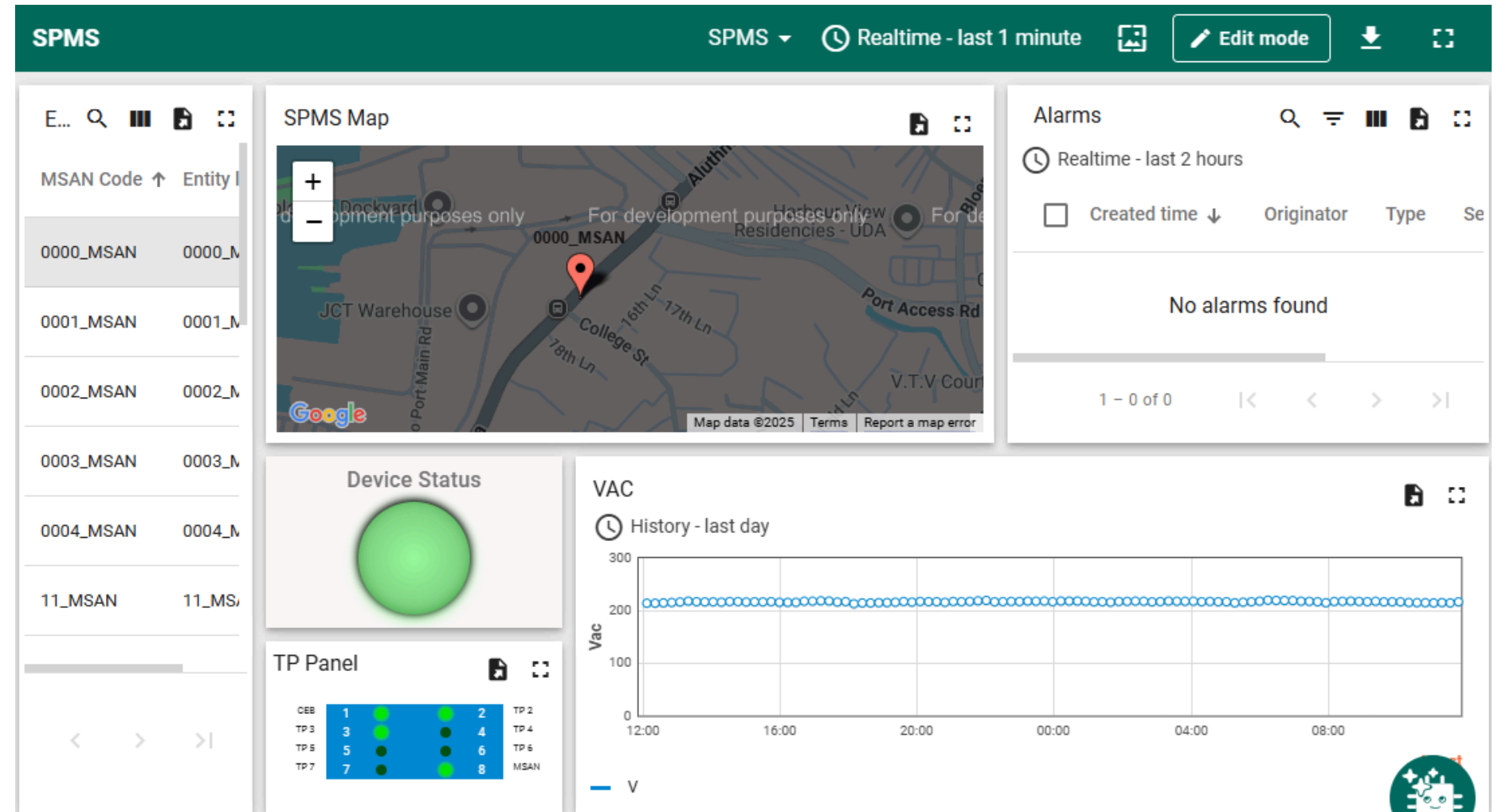
IoT Device



Technical Specifications

8 Monitoring Probes
12V Power Input
Connectivity - Mobitel 4G

Dashboard



Breaker Status Grid
Device Status (Online)
Live voltage graph

Expected Outcomes

Cost Reduction

Significantly reduce unnecessary field visits.

Improved Efficiency

Save time and resources for maintenance teams.

Enhanced Reliability

Boost MSAN system uptime and maintenance effectiveness.

Project Objectives

Power Status

Detect 230V AC presence

Log Voltage Fluctuations

Identify and log abnormal readings.

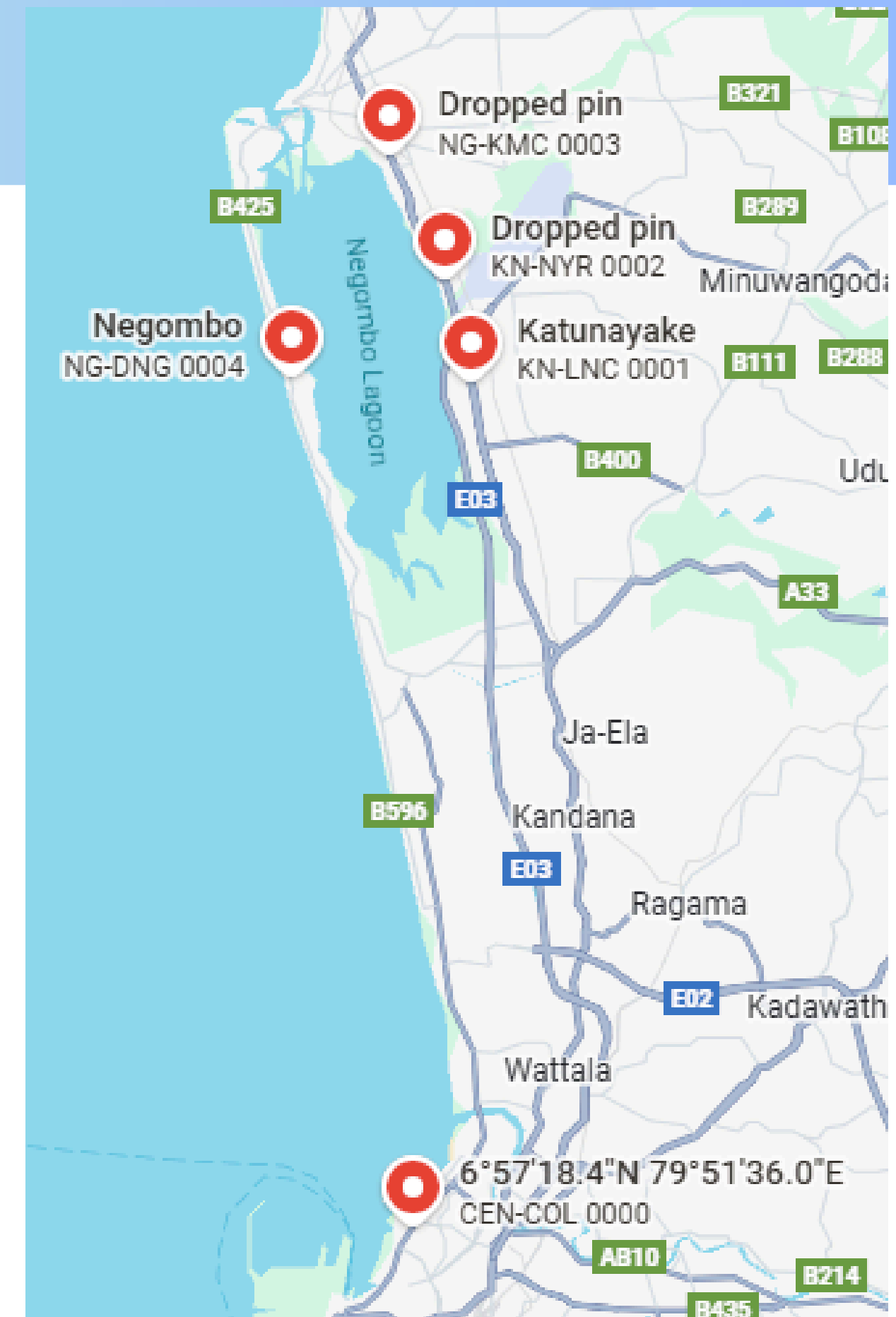
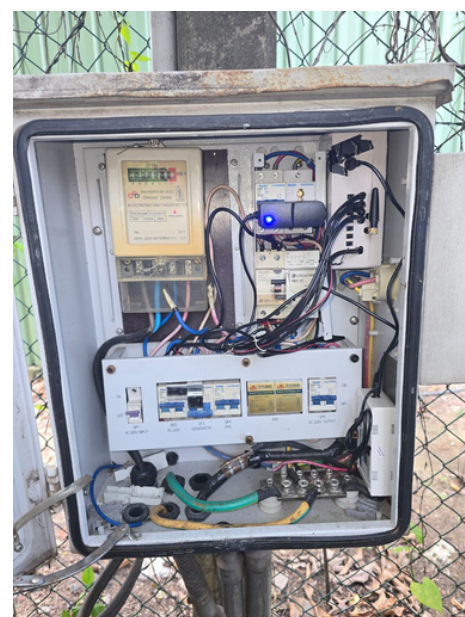
Alarm Correlation

Compare IoT events with MSAN alarm logs

Pilot Test

Total 5 devices are installed and in operation

No	Device Name	Region	MSAN	Install Date
1	0000	Central	CEN - COL	10/15/2025
2	0001	Katunayake	KN - LNC	11/12/2025
3	0002		KN - NYR	11/12/2025
4	0003	Negambo	NG - KMC	11/14/2025
5	0004		NG - DNG	11/14/2025



Testing Process

Goal:

Ensure IoT readings and MSAN alarms show consistent behavior during real-time power events.

- IoT recorded voltage data.
- MSAN generated network alarms.
- Datasets compared for correlation.

NOC Alarm Data

datetime	node	reason	alstatus	c
11/5/2025 14:21	CEN-COL_MSAG5200_01	AC Mains Failure	cleared	
11/5/2025 14:18	CEN-COL_MSAG5200_01	AC Mains Failure	new	
11/3/2025 14:10	CEN-COL_MSAG5200_01	The link between the se	cleared	
11/3/2025 14:11	CEN-COL_MSAG5200_01	The link between the se	new	
10/29/2025 10:40	CEN-COL_MSAG5200_01	AC Mains Failure	cleared	
10/29/2025 10:39	CEN-COL_MSAG5200_01	AC Mains Failure	new	
10/29/2025 10:31	CEN-COL_MSAG5200_01	AC Mains Failure	cleared	
10/29/2025 10:29	CEN-COL_MSAG5200_01	AC Mains Failure	new	
10/28/2025 14:07	CEN-COL_MSAG5200_01	Cabinet Door Opened:	cleared	
10/28/2025 14:05	CEN-COL_MSAG5200_01	Cabinet Door Opened:	new	
10/28/2025 13:40	CEN-COL_MSAG5200_01	Cabinet Door Opened:	cleared	
10/28/2025 13:40	CEN-COL_MSAG5200_01	Cabinet Door Opened:	new	
10/28/2025 13:41	CEN-COL_MSAG5200_01	Cabinet Door Opened:	cleared	
10/28/2025 13:31	CEN-COL_MSAG5200_01	Cabinet Door Opened:	new	

Scenario 1

This alarm is generated because our team is doing some changes of the IoT system and had to turn off the main power & IoT system

For that reason we will not use this as a test case

Scenario 1

NOC Alarms

11/5/2025 2:18:20 PM



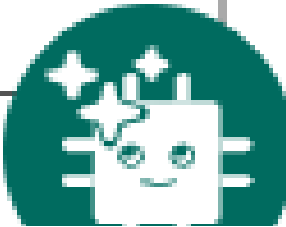
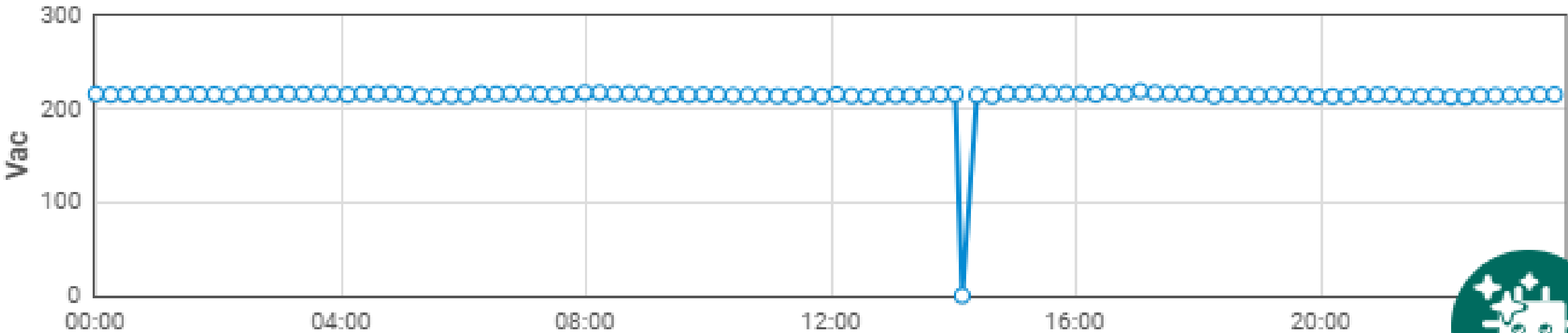
IoT Alarms

11/5/2025 2:08:11PM

VAC



History - from 2025-11-04 23:59:00 to 2025-11-05 23:59:00



Scenario 2

NOC Alarms

No NOC Alarm



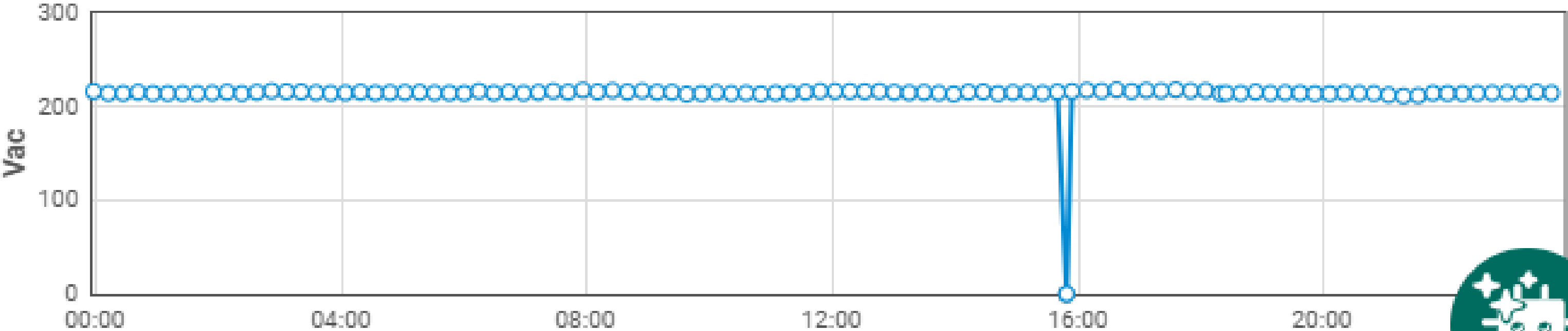
IoT Alarms

11/7/2025 3:48:50PM

VAC



History - from 2025-11-06 23:56:00 to 2025-11-07 23:56:00



Conclusion

NOC Alarms

11/5/2025 2:18:20 PM

No NOC Alarm

IoT Alarms

11/5/2025 2:08:11PM

11/7/2025 3:48:50PM

- **1 NOC & 2 IoT** Alarms within period of 27 days
- Scenario 1 matched with both systems.
- Scenario 2 **Only IoT system** report the power failure

According to above data we can come to the conclusion of that this IoT system can detect power failures with satisfactory accuracy for the use of SLT